

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T10-43-42-913Z

Run ID: #2

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Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming through the induction of pluripotency holds significant promise for regenerative medicine, allowing for the transformation of somatic cells into induced pluripotent stem cells (iPSCs). The objective of this experiment, identified as Run ID #2, was to evaluate the efficacy of OSKM transfection in reprogramming somatic cells under oncogenic stress conditions.

2. Materials and Methods

The experiment employed repeated transfection of somatic cells with Yamanaka factors (OSKM: Oct4, Sox2, Klf4, and c-Myc) under oncogenic stress to induce pluripotency. Computational analysis using scVI was planned but not conducted during this run, focusing instead on observing the population dynamics post-intervention.

Experimental Interventions:

- Oncogenic Stress Induced
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

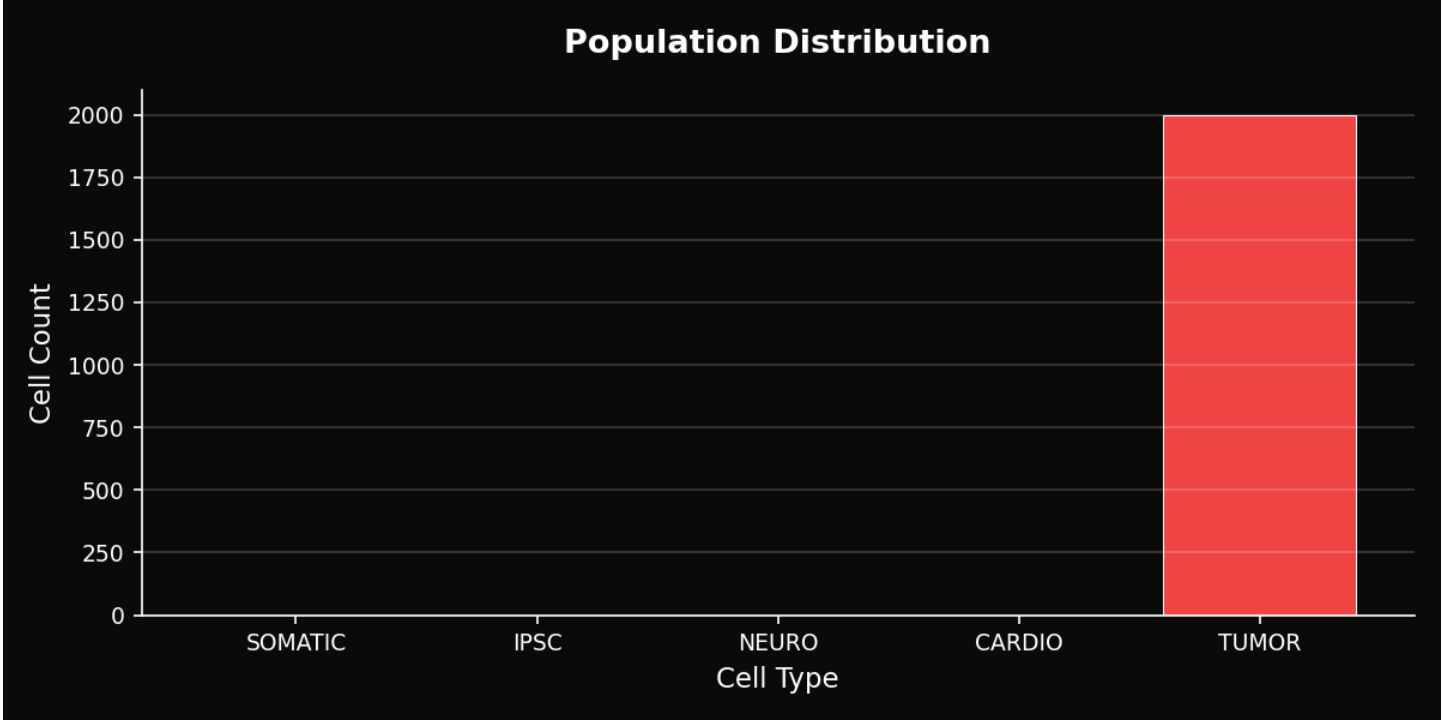
3. Results

Post-intervention analysis revealed a lack of successful reprogramming, with the final population comprising 2000 tumor cells and no detectable somatic, iPSC, neurogenic, or cardiogenic cell types. Despite multiple rounds of OSKM transfection, the oncogenic stress appeared to dominate the cellular landscape, preventing the emergence of desired pluripotent or differentiated cell types.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	0	0.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	2000	100.0%

4. Discussion

The findings suggest that under the given conditions, oncogenic stress may inhibit the reprogramming capacity of OSKM factors, favoring tumorigenesis over pluripotency. This outcome highlights the critical need to balance oncogenic signals when attempting cellular reprogramming, as excessive stress may predispose cells towards malignant transformation. Future studies should consider modulating the oncogenic environment or employing additional safeguards to improve reprogramming efficiency while minimizing tumor risk.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.